

Veil: Private Communication Through Entangled Relay

A Blockchain-Free Protocol for Metadata-Resistant Messaging
via Proof-of-Relay and Social Sybil Resistance

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Abstract

We present Veil, a decentralized messaging protocol that unifies metadata protection, spam prevention, and offline message delivery through a single mechanism: *Proof-of-Relay*. In Veil, sending a message requires a zero-knowledge proof that the sender has faithfully relayed messages for others through a stratified mixnet. The relay work itself constitutes the anonymizing infrastructure, eliminating the need for cryptocurrency tokens, blockchain consensus, or trusted third parties.

We make three contributions. First, we prove that *bilateral non-transferable credits* with epoch-bound nullifiers achieve incentive compatibility without global state—a general result applicable beyond messaging to any peer-to-peer system requiring fair exchange. Second, we establish a *Growth-Isolation Impossibility* theorem showing that no CRDT merge function can simultaneously resist inflation and guarantee completeness for monotonically growing verifiable evidence, and present a resolution via penalty-log CRDTs with locally-computed growth. Third, we prove a *constructive adversary bound*: any adversary controlling fraction f of relay nodes necessarily contributes sender anonymity entropy $H \geq \log_2(f \cdot N \cdot (1 - f^{L-1}))$ bits (via mandatory relay work) while individual deanonymization probability remains bounded by $\binom{K}{T} \cdot f^{L \cdot T} + O(1/N_l)$, ensuring that adversarial participation requires productive contribution while individual targeting remains negligible.

Veil requires no economic investment to participate—privacy is earned through device contribution alone. We analyze the protocol’s security under a global passive adversary with formal indistinguishability definitions, bound Sybil infiltration under depth-limited social vouching, and demonstrate mobile feasibility with verified constraint counts via Nova folding over BabyJubjub.

1 Introduction

Private electronic communication is a foundational requirement for free societies. An estimated 4.5 billion people live under governments that conduct mass surveillance of digital communications [29]. Journalists, dissidents, whistleblowers, and ordinary citizens require messaging systems that protect not only message *content* (solved by end-to-end encryption [2]) but also *metadata*—who communicates with whom, when, and how often [3].

The consequences of metadata exposure are concrete and severe. A journalist in Myanmar communicating with a source inside the military junta risks both lives if the communication pattern is observed—even when content is encrypted. An LGBTQ+ activist in Uganda, where homosexuality carries life imprisonment, cannot safely use any messaging system whose servers log contact graphs. A labor organizer in Saudi Arabia faces imprisonment if communication patterns with international press are identified. For these users, metadata *is* the content.

Existing approaches each solve part of this problem but leave critical gaps:

- **Signal** [2] provides strong content encryption but relies on centralized servers that observe metadata—and can be compelled to share it.
- **Nym** [4] provides metadata protection via incentivized mixnets but requires cryptocurrency participation—inaccessible in jurisdictions where crypto is banned (China, Algeria, Bangladesh) or where exchange access requires KYC documentation that 1.4 billion unbanked adults [30] cannot provide.
- **Tor** [5] provides low-latency anonymity but is vulnerable to traffic correlation by global adversaries, offers no incentive for relay operators, and is actively blocked by state-level DPI in Iran, China, and Russia.
- **SimpleX** [6] eliminates user identifiers but depends on relay servers and provides no spam resistance mechanism.

No system simultaneously achieves: (1) metadata resistance against a global passive adversary, (2) reliable offline delivery without trusted infrastructure, (3) spam prevention without identity or economic barriers, and (4) true decentralization without blockchain consensus. We identify that these four problems share a common root: *who stores and forwards messages, and why would they do it honestly?*

1.1 The Core Insight

Bitcoin [1] solved three problems (double-spending, Sybil resistance, Byzantine fault tolerance) through one mechanism: proof-of-work combined with the longest-chain rule. We observe an analogous opportunity for private messaging.

In Veil, sending a message requires a zero-knowledge proof that the sender has faithfully relayed messages for others. The relay work simultaneously provides:

1. **Metadata protection:** Relayed messages traverse a stratified mixnet with Poisson delays—the relay infrastructure IS the anonymity network.
2. **Offline delivery:** Nodes are incentivized to store messages for offline recipients because storage earns relay credits.
3. **Spam prevention:** Sending is metered by verifiable relay contribution—spamming requires proportional service to the network.
4. **Sybil cost:** Earning credits requires relaying for reputable nodes, which requires social attestation to join—creating a time-cost for adversarial infiltration.

Unlike proof-of-work, proof-of-relay is *useful work*—the computation directly produces the system’s utility. Unlike token-based incentives, relay credits are non-transferable, locally-verifiable, and require no global ledger.

We propose the *constructive adversary principle* as a design heuristic: in any system where adversarial participation requires productive work for honest users, the adversary’s investment simultaneously benefits the network. We formalize this for mixnet relay and leave cross-domain validation to future work.

1.2 Contributions

1. **Proof-of-Relay Protocol** (§4): A complete messaging protocol where a single mechanism unifies transport, spam prevention, metadata protection, and incentive alignment without tokens or blockchain.
2. **Bilateral Credit Incentive Theorem** (§5): We prove that non-transferable, locally-verifiable, epoch-bound bilateral credits achieve incentive compatibility as a dominant strategy equilibrium without global state—resolving the nullifier reuse problem via verifier-binding without consensus.
3. **Growth-Isolation Impossibility and Resolution** (§6): We prove that no CRDT merge function simultaneously provides inflation resistance and completeness for growing verifi-

able evidence, covering all merge strategies including authenticated increment chains. We present a penalty-log CRDT that resolves this impossibility—a construction with applications to decentralized reputation, distributed moderation, and peer-to-peer trust beyond messaging.

4. **Constructive Adversary Bound** (§7): We prove, with formal Shannon entropy quantification and hypergeometric correlation correction, that any adversary in Veil provides more aggregate privacy than it can destroy for any individual target.

2 Formal Security Definitions

We define the security properties Veil targets via standard indistinguishability games. These definitions follow the AnoA framework [22] adapted to our setting. Together, Definitions 1–3 formalize the informal goal of “metadata protection”: sender anonymity hides *who* sent a message, unlinkability hides *who talks to whom*, and unobservability hides *whether* a message was sent at all.

Definition 1 (Sender Anonymity). *Let $\mathcal{A}$ be a PPT adversary. For any two honest senders $s_0, s_1 \in V$, any honest recipient r , and any message m , define:*

$$\text{Adv}_{\mathcal{A}}^{SA} = |\Pr[\mathcal{A}(\text{View}_{\mathcal{A}}(s_0 \rightarrow r, m)) = 1] - \Pr[\mathcal{A}(\text{View}_{\mathcal{A}}(s_1 \rightarrow r, m)) = 1]|$$

where $\text{View}_{\mathcal{A}}$ includes all network observations, controlled node states, and timing information available to $\mathcal{A}$. Veil satisfies sender anonymity if $\text{Adv}_{\mathcal{A}}^{SA} \leq \text{negl}(\kappa)$ for all PPT $\mathcal{A}$ controlling $f < f_{\max}$ nodes.

Definition 2 (Sender-Receiver Unlinkability). *For messages m_1, m_2 from senders s_1, s_2 to recipients r_1, r_2 :*

$$\text{View}_{\mathcal{A}}(s_1 \rightarrow r_1, s_2 \rightarrow r_2) \approx_c \text{View}_{\mathcal{A}}(s_1 \rightarrow r_2, s_2 \rightarrow r_1)$$

where $\approx_c$ denotes computational indistinguishability.

Definition 3 (Sender Unobservability). *For any honest sender s and any two strategies σ_{send} (“ s sends message m to r ”) and σ_{idle} (“ s sends nothing”):*

$$\text{Adv}_{\mathcal{A}}^{UO} = |\Pr[\mathcal{A}(\text{View}_{\mathcal{A}}(\sigma_{\text{send}})) = 1] - \Pr[\mathcal{A}(\text{View}_{\mathcal{A}}(\sigma_{\text{idle}})) = 1]| \leq \text{negl}(\kappa)$$

Definition 4 (Relay Credit Soundness). *No PPT adversary $\mathcal{A}$ can produce a valid relay credit proof π accepted by an honest verifier without having performed N_{credit} relay operations for distinct reputable nodes, except with probability $\text{negl}(\kappa)$.*

3 System Model and Threat Model

3.1 Network Model

The Veil network consists of N nodes. Every node is simultaneously a user and a relay—there is no client/server distinction. Nodes have intermittent connectivity (modeling mobile devices) and communicate via authenticated channels.

Definition 5 (Veil Network). *A Veil network $\mathcal{N} = (V, E, \mathcal{R}, \mathcal{C})$ consists of:*

- V : Set of N nodes, each possessing a long-term key pair (sk_i, pk_i) over BabyJubjub.
- E : Communication links (unreliable, asynchronous).
- $\mathcal{R}$: Reputation state, maintained as a distributed penalty-log CRDT.
- $\mathcal{C}$: Credit state, maintained locally at each node.

Nodes are organized into a *stratified topology* with L layers, following Loopix [8]. Each message traverses one randomly-selected node per layer.

3.2 Threat Model

Definition 6 (Global Passive Adversary). *The adversary $\mathcal{A}$ can observe all network links simultaneously, recording packet timing, size, and endpoints. $\mathcal{A}$ additionally controls a fraction $f < f_{\max} = 0.22$ of nodes (active corruption).*

The bound $f_{\max} = 0.22$ derives from the penalty detection threshold $\tau = 0.05$ via the coupling $f_{\max} < \sqrt{\tau}$ (see Proposition 8).

Adversary capabilities:

- Observe all network traffic (timing, volume, endpoints)
- Compromise up to fraction $f < 0.22$ of nodes (Byzantine behavior)
- Create Sybil identities (bounded by social attestation mechanism)
- Store historical network observations indefinitely
- Collude among controlled nodes and coordinate strategy

Adversary limitations:

- Cannot break the discrete logarithm assumption over BabyJubjub
- Cannot forge zk-SNARK proofs (computational soundness)
- Social infiltration has irreversible cost (see below)

Social infiltration cost model: We do not assume social relationships are unforgeable—intelligence agencies cultivate relationships professionally. Rather, we bound the *activation cost*. A seed node that vouches for k eventual Sybils incurs reputation contagion ($\gamma_{\text{contagion}} = 0.5$ per activated Sybil penalty). Activating n Sybil nodes burns $\lceil n/B_{\max} \rceil$ seed nodes below $R_{\min}$ (permanently unusable). The adversary faces an irreversible choice: maintain long-term infiltration capability OR execute a one-time attack that destroys all invested social capital. This mirrors the tradeoff intelligence agencies face with human sources—burning a source is a non-recoverable strategic loss.

Mass vs. targeted surveillance: The bound $f_{\max} = 0.22$ provides robustness against mass-surveillance infrastructure, where adversarial nodes are distributed across the population and observe traffic in bulk. It does NOT guarantee protection against targeted infiltration of a specific user’s social neighborhood. A nation-state adversary dedicating resources to one target may concentrate controlled nodes around that target’s relay paths. Such targeted attacks are bounded by the social attestation mechanism’s time-cost and reputation-contagion risk (Corollary 12), not by $f_{\max}$ alone. This mirrors Tor’s operational reality: strong against mass observation, diminishing guarantees against sustained targeted surveillance of a single individual.

Explicit non-goals:

- Real-time chat (we accept 30–120 second delivery latency)
- Anonymity against an adversary controlling $> f_{\max}$ of reputable nodes
- Deniability of network participation (see §10.2 for network-level blocking resistance)
- Protection against compromised endpoint devices

4 Protocol Design

4.1 Overview

Veil’s protocol consists of five interlocking components:

1. **Mixnet Relay** (§4.2): Stratified mixnet with Poisson delays and cover traffic.
2. **Proof-of-Relay Credits** (§4.3): Zero-knowledge proofs of relay work, earned bilaterally and verified locally.
3. **Proof-of-Entanglement** (§4.4): Social Sybil resistance via depth-limited vouching with reputation contagion.
4. **Threshold Delivery** (§4.6): Message fragmentation and staggered emission for offline delivery and path-diversity security.

5. **Bootstrap Protocol** (§4.5): Mechanism for new nodes to enter the network without deadlock.

4.2 Mixnet Architecture

4.2.1 Stratified Topology

Nodes are assigned to $L = 3$ layers based on reputation and uptime, rotating every epoch $\mathcal{E}$ (24 hours). Messages traverse one randomly-selected node per layer. Within each layer, path selection uses uniform random sampling independent of other shares and other messages.

4.2.2 Packet Format

All packets are fixed-size (2048 bytes), constructed using a modified Sphinx [9] format with circuit-friendly primitives:

- **Key exchange:** ECDH over BabyJubjub
- **Header encryption:** Poseidon-CTR [10] (header only, ~ 400 bytes per hop)
- **Payload encryption:** Stream cipher outside SNARK (relay proves correct routing, not payload re-encryption)
- **MAC:** Poseidon-MAC
- **Routing/hashing:** Poseidon hash

The circuit proves correct Sphinx header processing (routing computation and header transformation). Payload is encrypted using a symmetric stream cipher derived from the shared ECDH secret; the relay forwards the re-encrypted payload without proving this step inside the SNARK.

4.2.3 Constraint Analysis

Per-relay-step circuit size (header-only Poseidon-CTR):

Table 1: R1CS constraint breakdown per relay step (estimated vs. measured)

Component	Estimate	Measured	Note
ECDH (variable-base scalar mult)	6,500	3,982	Complete twisted Edwards formulas
Poseidon-CTR header (13 blocks)	3,900	3,120	13×240 (partial-round opt.)
Poseidon-MAC (2 invocations)	600	480	2×240
Poseidon routing hash (3 invocations)	900	720	3×240
Receipt verification (1 Poseidon)	300	240	
Merkle membership (depth-14)	4,200	3,403	14×243 (incl. selector)
Glue (wiring, range checks)	—	442	Boolean/equality constraints
Total per relay step	16,400	12,387	−25% vs. estimate

Our reference implementation on arkworks 0.4 [34] achieves 12,387 R1CS constraints per relay step—25% below the conservative estimate—due to optimized twisted Edwards complete addition formulas and partial-round compression in the Poseidon gadget. Extrapolating linearly from constraint count, we estimate $\sim 150\text{ms}$ per Nova fold on ARM Cortex-A53 ($\pm 50\%$ depending on memory access patterns and commitment scheme overhead; actual timing may range 100–225ms). Hardware benchmarks are pending but the constraint reduction provides substantial margin below the 200ms feasibility threshold.

With $N_{\text{credit}} = 8$ relay operations folded via Nova IVC, the per-fold circuit contains $\sim 12,400$ constraints. The final Groth16 compression proves the entire folded chain. The circuit proves

correct Sphinx header processing (routing computation and header transformation); payload encryption uses a stream cipher outside the SNARK.

4.2.4 Poisson Mixing Delays

Each relay node delays each packet by an independent sample from an exponential distribution:

$$\text{delay}_l \sim \text{Exp}(\mu_l), \quad \mu_1 = 500\text{ms}, \mu_2 = 1000\text{ms}, \mu_3 = 500\text{ms} \quad (1)$$

Expected total path delay: 2 seconds. The 95th percentile delivery time is approximately 6 seconds per share, with total message delivery (including staggered emission) in the range of 30–120 seconds.

4.2.5 Cover Traffic

Every node generates three types of cover traffic at configured Poisson rates:

- **Loop cover** ($\lambda_{\text{loop}} = 1/10\text{s}$): Self-addressed packets for active attack detection.
- **Drop cover** ($\lambda_{\text{drop}} = 1/15\text{s}$): Random-destination packets dropped at final hop.
- **Payload cover** ($\lambda_{\text{payload}} = 1/20\text{s}$): Indistinguishable from real traffic.

Total cover emission rate: $\lambda_{\text{total}} = 0.1 + 0.067 + 0.05 = 0.217$ packets/second.

Daily cover bandwidth (sending only): $0.217 \times 86400 \times 2048 \approx 38.4$ MB/day. Under the adaptive $\lambda/4$ mode for mobile devices: ~ 9.6 MB/day (cover sending only; total including relay obligations: 15–18 MB/day, see §8.5).

Mobile profile security trade-off: The adaptive $\lambda/4$ mode reduces cover emission to $\lambda_{\text{total}} \approx 0.054$ pkt/s. This proportionally increases the observation window a GPA needs for statistical disclosure: at full rate, an adversary accumulates $0.217 \times T$ observations in time T ; at $\lambda/4$, the same accumulation requires $4T$. Theorem 9 holds with the substitution $\lambda \rightarrow \lambda/4$ —the asymptotic guarantee is unchanged, but the concrete mixing window within which a single send is hidden expands by factor 4 (i.e., the anonymity set refreshes more slowly). Full-rate nodes dominate the network’s anonymity contribution; mobile nodes benefit from this set while contributing proportionally less.

Real message shares are emitted at the same Poisson rate as cover traffic, providing sender unobservability (Definition 3). When a node has no message to send, it emits cover; when sending, a cover slot is replaced with a real share—indistinguishable to any observer.

4.3 Proof-of-Relay Credits

4.3.1 Credit Mechanism

To send one message, a node must prove it has relayed $N_{\text{credit}} = 8$ packets for other reputable nodes within the last $W = 2$ *credit windows*.

Definition 7 (Time Parameters). *We distinguish two time scales:*

- **Epoch** ($\mathcal{E} = 24$ hours): *Layer rotation and reputation snapshot period.*
- **Credit window** ($W_{\text{credit}} = 24$ days): *Duration for which relay receipts remain valid for credit proofs. Credits are valid for at most 2 credit windows (= 48 days).*

Credit window calibration. The 24-day window is sized for mobile usage patterns. In a network of $N = 5,000$ nodes with 3 layers, each layer contains $\sim 1,667$ nodes. A Layer-1 node receives relay traffic from all senders whose path includes it—at the full cover rate, approximately $N \times \lambda_{\text{total}}/N_l \approx 5000 \times 0.217/1667 \approx 0.65$ packets/second. A mobile node online for 5 minutes receives ~ 195 packets, far exceeding the $N_{\text{credit}} = 8$ threshold. Even at 10% network participation (only 500 of 5,000 nodes online), per-node relay rate drops to ~ 0.065 pkt/s, yielding ~ 19.5 packets in 5 minutes; $P(\text{Poisson}(19.5) \geq 8) > 0.999$. The 24-day window

provides ample margin even for nodes with very intermittent connectivity (e.g., 5 minutes/day, 3 days/week), while remaining short enough to bound stale-credit accumulation. (This calculation assumes uniform traffic distribution across layers; in practice, per-layer load varies with node count and reputation distribution, but the $24\times$ margin over the minimum makes the conclusion robust to moderate imbalance.)

Definition 8 (Relay Credit). *A relay credit is a zero-knowledge proof π_{relay} attesting:*

“I have correctly processed N_{credit} Sphinx packet headers for distinct reputable nodes within the last $2 \cdot W_{\text{credit}}$ days, receiving valid acknowledgments for each, where each receipt issuer is a member of the current epoch’s reputable set.”

Without revealing: which packets, which nodes, or exact timing.

4.3.2 Bilateral Receipts

When node A relays a packet to next-hop node B :

1. B computes: receipt = Poseidon(packet_hash|| sk_B ||epoch)
2. B sends receipt to A (piggybacked on next communication)
3. A stores the receipt locally as evidence of relay work

4.3.3 Epoch-Bound Nullifiers with Verifier Binding

Each credit proof includes a nullifier that prevents reuse:

$$\text{nullifier} = \text{Poseidon}(sk_{\text{sender}}||\text{epoch}||\text{verifier_pk}||\text{nonce}) \quad (2)$$

The nullifier commits to: (1) the sender’s secret key, (2) the current epoch, (3) the specific first-hop verifier’s public key, and (4) a one-time nonce derived deterministically from the credit set.

Proposition 1 (No Double-Use Without Consensus). *The epoch-bound verifier-binding construction eliminates credit reuse across verifiers:*

1. **Same verifier:** Nullifier collision detected (verifier stores used nullifiers locally).
2. **Different verifier:** The SNARK circuit enforces that the nullifier’s verifier_pk field matches the actual first-hop node receiving the proof. A proof generated for verifier V is invalid at verifier $V' \neq V$ because the circuit checks $\text{verifier_pk} = pk_V$ against the verifier’s known public key.
3. **Different epoch:** Nullifier includes epoch; proofs from expired epochs are rejected.

Proof. The SNARK circuit takes as public input the verifier’s public key pk_V and the current epoch e . Inside the circuit, the prover demonstrates:

- Knowledge of sk such that (sk, pk_{sender}) is a valid key pair.
- The nullifier is correctly computed as $\text{Poseidon}(sk||e||pk_V||\text{nonce})$.
- The nonce is deterministically derived from the credit set being consumed.

Since pk_V is a public input verified by the receiving node (it checks: “is this MY public key?”), the same proof cannot validate at a different node V' (which would check against $pk_{V'}$). The circuit’s soundness guarantees that no PPT prover can produce a valid proof with a different pk_V embedded while passing verification at V' . $\square$ $\square$

Privacy implication: Verifier-binding does NOT compromise sender anonymity. The first-hop relay already knows it is interacting with the sender (by definition—it receives the message directly). The sender’s identity is protected from all subsequent hops by onion encryption. Binding the credit to the first hop reveals no additional information beyond what the first hop already observes.

4.3.4 Incremental Proving via Nova Folding

Rather than proving all N_{credit} relay operations in a single circuit, we employ Incrementally Verifiable Computation (IVC) via Nova folding [11]:

1. Each relay operation produces a step witness w_i (packet hash, receipt, Merkle proof of issuer reputation).
2. The witness is folded into a running proof: $\pi_i = \text{Fold}(\pi_{i-1}, w_i)$.
3. When sending, the running proof is compressed into a final Groth16 SNARK with verifier-binding. The final compression is pre-computed and cached after the N_{credit} -th fold completes—it is NOT generated at send time, preventing timing correlation between proof generation and message emission.

Per-fold cost: $\sim 200\text{ms}$ on mobile hardware (accounting for Merkle proof overhead). Final compression: ~ 2 seconds (performed once, cached until consumed).

4.3.5 Reputation-Checked Receipts

To prevent self-relay and collusion attacks, the relay proof circuit includes a Merkle membership proof verifying that each receipt issuer belongs to the current epoch’s *reputable set*:

Definition 9 (Reputable Set). *At each epoch e , the reputable set $\mathcal{RS}_e \subset V$ contains all nodes with reputation $R > R_{\min}$. The set is published as a Merkle tree root root_e (public parameter, updated each epoch via the penalty-log CRDT).*

The relay proof circuit checks, for each receipt r_i from node B_i :

1. Receipt is well-formed: $r_i = \text{Poseidon}(\text{packet_hash}_i \| sk_{B_i} \| e_i)$
2. Issuer is reputable: $\text{MerkleVerify}(pk_{B_i}, \text{path}_i, \text{root}_{e_i}) = 1$
3. Distinctness: All N_{credit} receipt issuers have distinct public keys

This adds $\sim 4,200$ constraints per receipt (14-level Poseidon Merkle proof for $N \leq 16,384$ nodes), included in the constraint analysis of Table 1.

4.3.6 Credit Properties

Credits are:

- **Bilateral**: Produced by interaction between exactly two parties.
- **Non-transferable**: Bound to prover’s secret key via the SNARK.
- **Locally verifiable**: Any node can verify a SNARK proof given only public parameters.
- **One-time-use**: Verifier-bound nullifier prevents reuse at any node (Proposition 1).
- **Ephemeral**: Credits expire after $2 \cdot W_{\text{credit}}$ days, preventing accumulation.
- **Reputation-gated**: Receipt issuers must be in the reputable set (no self-relay via Sybils).

4.4 Proof-of-Entanglement: Social Sybil Resistance

4.4.1 Joining the Network

To participate as a relay node, a new node N requires vouching from $K_{\text{vouch}} \geq 3$ existing nodes with reputation $R > R_{\min}$.

Definition 10 (Vouching Protocol). *Voucher V_i produces:*

1. *A vouching commitment: $C_i = \text{Commit}(pk_N, pk_{V_i}, \text{epoch}, \text{nonce})$*
2. *A zero-knowledge proof π_{vouch} : “My reputation $R > R_{\min}$ AND I have not exceeded my vouching budget this credit window.”*

Both are submitted to the distributed penalty-log CRDT.

4.4.2 Depth-Limited Vouching

Each node with reputation R may vouch at most $B = \lfloor R \times 3 \rfloor$ new nodes per *credit window* (24 days). Critically, **vouched nodes cannot vouch others** (vouching depth = 2). This bounds Sybil creation:

Proposition 2 (Linear Sybil Bound). *Given s seed adversarial nodes with maximum reputation, the maximum number of Sybil nodes created after t credit windows is:*

$$S(s, t) = s \times B_{\max} \times t \quad (3)$$

Growth is linear in time, not exponential. For $s = 10$, $B_{\max} = 3$: at most 30 new Sybils per credit window (24 days).

Proof. Each seed vouches $B_{\max}$ nodes per credit window. Vouched nodes have depth 2 and cannot vouch. Therefore total Sybils after t credit windows = $s \times B_{\max} \times t$. No branching occurs beyond depth 2. $\square$ $\square$

4.4.3 Reputation Contagion

When a vouched node N is penalized, a *contagion penalty event* is propagated via the CRDT:

1. N 's reputation: direct penalty event with severity $1 - \gamma_{\text{direct}}$, where $\gamma_{\text{direct}} = 0.1$
2. Each voucher V_i : contagion penalty event with severity $1 - \gamma_{\text{contagion}}$, where $\gamma_{\text{contagion}} = 0.5$

The contagion penalty is a proper penalty event propagated through the CRDT (set union of penalty events), consistent with the penalty-log architecture where all reputation decreases propagate as penalty events and growth is computed locally (see Definition 14 for the formal reputation computation).

4.5 Bootstrap Protocol

New nodes face a cold-start problem: they need relay credits to send but cannot earn credits without being assigned to relay layers (which requires reputation). We resolve this via a two-phase bootstrap:

Definition 11 (Bootstrap Protocol). *A new node N entering the network proceeds in two phases:*

1. **Training phase:** N is assigned to a dedicated Layer 0 (training layer) for the first $W_{\text{bootstrap}} = 500$ relay operations (≈ 2 –3 days at typical traffic rates). Traffic through Layer 0 has weaker anonymity guarantees (smaller anonymity set) but allows N to earn initial credits. During this phase, N may send up to $C_{\text{bootstrap}} = 2$ messages using bootstrap credits (free credits granted upon successful vouching).
2. **Graduation:** After $W_{\text{bootstrap}}$ verified relay operations, N is eligible for assignment to Layers 1–3 based on accumulated reputation.

Incentive for routing through Layer 0: Existing nodes route a fraction $\rho = 0.1$ of their *drop cover traffic* through Layer 0 nodes. This cover traffic provides relay opportunities for new nodes at zero privacy cost (drop cover is discarded at the final hop regardless). The routing fraction ρ is individually rational: it increases network capacity by bootstrapping new relays, benefiting all nodes in future epochs.

4.6 Threshold Delivery and Offline Storage

4.6.1 Message Fragmentation

Message M is split into $K = 5$ shares via Shamir's Secret Sharing over a finite field, with threshold $T = 3$. Any T shares reconstruct M ; fewer than T reveal no information (information-theoretic security).

4.6.2 Staggered Emission

Shares are emitted at the same Poisson rate as cover traffic:

$$t_{\text{emit}}(\text{share}_i) = t_{\text{base}} + \sum_{j=1}^i \text{Exp}(\lambda_{\text{cover}}) \quad (4)$$

This ensures that real message emission is statistically indistinguishable from ongoing cover traffic, providing sender unobservability (Definition 3).

4.6.3 Anonymous Mailboxes

Each recipient maintains $D = 5$ independent mailbox queues, each hosted on a different relay node and identified by a random, unlinkable queue-ID shared only with the sender (per-conversation).

Recipients retrieve from all D mailboxes at fixed intervals ($T_{\text{retrieve}} = 30$ seconds), downloading fixed-size blocks (padded if empty). This constant-rate retrieval prevents timing correlation between message deposit and retrieval.

4.6.4 Storage Incentive

Mailbox nodes earn credits for storing shares: $\text{credit} = \text{duration}/6\text{h}$. Maximum storage duration: 7 days.

5 Bilateral Credit Incentive Compatibility

We prove that Veil’s credit system achieves incentive compatibility without global consensus. Our proof proceeds in layers, with each layer’s correctness independent of the others, resolving the circular dependency that would arise from a monolithic argument.

5.1 Layered Proof Architecture

The incentive analysis decomposes into four independent layers:

1. **Layer 0 (CRDT Correctness):** The penalty-log CRDT’s merge operation (set union) is correct by construction—it satisfies commutativity, associativity, and idempotency regardless of node behavior. This is axiomatic: set union works even if nodes gossip maliciously (they can delay or withhold, but cannot corrupt the merge).
2. **Layer 1 (Gossip Incentive Compatibility):** Given correct CRDT semantics, honest gossip of penalty events is individually rational (§5.5).
3. **Layer 2 (Penalty Detection):** Given honest gossip, the statistical drop-rate mechanism detects misbehavior (§5.3).
4. **Layer 3 (Relay Incentive Compatibility):** Given working penalty detection, honest relay is a dominant strategy (Theorem 4).

Each layer depends only on the one below it. Layer 0 has no dependencies (mathematical truth). We now prove each layer.

5.2 Layer 1: Gossip Incentive Compatibility

Lemma 3 (Gossip is Individually Rational). *In the penalty-log CRDT, honest propagation of penalty events is individually rational for all nodes, independent of relay incentives.*

Proof. Consider node i observing a penalty event p against node j . Node i 's options:

1. **Propagate p :** All neighbors learn of j 's misbehavior. Node j 's reputation decreases everywhere, reducing j 's assignment to relay layers. This benefits i because: (a) i is less likely to receive dropped packets from j in future, and (b) i 's relative reputation increases.
2. **Withhold p :** Node j retains reputation at nodes that haven't seen p from other paths. But this does NOT benefit i : j being reputable at other nodes doesn't help i (no transferable credits, no shared resource). Meanwhile, i still suffers from j 's misbehavior in future interactions.

Withholding p provides zero benefit to i (credits are non-transferable; i cannot profit from j 's reputation) while propagation provides positive benefit (reduced future interaction with misbehaving j). Therefore propagation strictly dominates withholding.

Colluding case: Even if i and j collude, i gains nothing by hiding j 's penalties— j 's reputation helps j , not i . Collusion helps only if credits were transferable (which they are not, by construction).

Cartel resistance: Consider a coalition $\mathcal{K}$ of k nodes that mutually suppress each other's penalties. This delays propagation but cannot prevent it: penalty events are generated by the *victim's* observation of dropped packets (the sending or receiving counterpart of a misbehaving relay). Since relay paths are assigned uniformly at random, each misbehaving cartel member interacts with a non-cartel (honest) counterpart with probability $\geq 1 - f$ per relay operation. After $W_{\min} = 500$ observations, honest counterparts independently generate the penalty event and propagate it through honest gossip paths—which reach 95% of the network within 5 rounds regardless of cartel withholding (verified empirically in §9). The cartel delays penalty propagation by at most ~ 1 gossip round but cannot suppress it. $\square$ $\square$

5.3 Layer 2: Penalty Detection

Given honest gossip (Layer 1), penalty events propagate reliably. The detection mechanism itself is analyzed in §6.4.

5.4 Game-Theoretic Model

Definition 12 (Relay Game). *The Relay Game $\Gamma = (V, \mathcal{S}, u)$ is an infinitely-repeated game where:*

- *Players:* $V = \{v_1, \dots, v_N\}$ (network nodes)
- *Strategy per round:* $s_i \in \{\text{relay-honestly}, \text{drop}, \text{free-ride}\}$
- *Utility:* $u_i = \alpha \cdot \text{messages_sent}_i - \beta \cdot \text{relay_cost}_i - \gamma \cdot \text{penalty_cost}_i$
- *Constraint:* $\text{messages_sent}_i \leq \lfloor \text{valid_credits}_i / N_{\text{credit}} \rfloor$

where α = value of sending one message, β = cost of one relay operation, γ = cost of reputation loss from penalty.

Definition 13 (Bilateral Non-Transferable Credit). *A credit c is bilateral non-transferable if it satisfies:*

1. **Bilateral:** c is produced by interaction between exactly two parties (A, B).
2. **Non-transferable:** c is cryptographically bound to A 's secret key; possession cannot be delegated.
3. **Locally-verifiable:** Any verifier can check c 's validity given only A 's proof and public parameters.
4. **One-time-use:** c is consumed upon use; verifier-bound nullifier prevents reuse at any verifier (Proposition 1).

5. **Reputation-gated:** c is only valid if the receipt issuer belongs to the reputable set at issuance epoch.

Theorem 4 (Bilateral Credits Achieve Dominant Strategy Equilibrium). *In the relay game Γ with bilateral non-transferable credits, honest relay is a dominant strategy—optimal regardless of other players’ strategies—without requiring global state agreement, provided:*

$$\frac{\alpha}{N_{\text{credit}}} > \beta \quad (5)$$

(i.e., the per-credit value of sending exceeds the cost of one relay operation).

Proof. We show that for any player v_i and any strategy profile σ_{-i} of other players, honest relay yields strictly higher utility than any alternative.

Step 1: No double-spending problem exists.

Credits are verifier-bound: a proof generated for first-hop node V is computationally infeasible to reuse at node $V' \neq V$ (Proposition 1). Credits are non-transferable: bound to sk_i via the SNARK circuit. Therefore:

- No two verifiers accept the same credit (verifier-binding).
- No two provers claim the same credit (key-binding).
- No expired credit is accepted (epoch-binding).

No race condition between competing claims exists. No global ordering is needed.

Step 2: Free-riding is strictly dominated.

A free-rider v_f that does not relay earns zero valid credits (receipts require actual relay interactions with reputable nodes, verified inside the SNARK via Merkle membership). With zero credits, v_f sends zero messages. Utility of free-riding:

$$u_f(\text{free-ride}) = \alpha \cdot 0 - \beta \cdot 0 = 0$$

Utility of honest relay (sending m messages over the credit window):

$$u_i(\text{honest}) = \alpha \cdot m - \beta \cdot m \cdot N_{\text{credit}} = m \cdot (\alpha - \beta \cdot N_{\text{credit}}) > 0$$

by the theorem’s condition. Therefore honest play strictly dominates free-riding regardless of σ_{-i} .

Step 3: Dropping is strictly dominated.

A node v_d that accepts packets but drops them:

1. Receives no valid receipt from the next hop (which never received the packet). Therefore, dropped packets generate no credits.
2. If v_d ’s drop rate exceeds τ over window $W_{\min}$ observations (Layer 2, Proposition 8): v_d is penalized, losing reputation and future relay assignment.

Even without detection (drop rate $< \tau$), dropping provides zero credits for dropped packets—the same as not relaying them. Combined with the risk of penalty ($\gamma > 0$):

$$u_d(\text{drop}) \leq u_d(\text{honest}) - \gamma \cdot P(\text{detected}) < u_d(\text{honest})$$

Dominant strategy conclusion: For every σ_{-i} , honest relay yields $u_i > 0$ while alternatives yield $u_i \leq 0$ (free-ride) or $u_i < u_i(\text{honest})$ (drop). This holds in every subgame of the repeated game. Therefore honest relay is a dominant strategy, and the profile (honest, ..., honest) is trivially also an SPNE. □ □

Corollary 5 (No Blockchain Required). *Since bilateral non-transferable credits with verifier-binding have no double-spending problem (Proposition 1), no race condition between competing claims exists. Global state agreement (blockchain, BFT consensus) is unnecessary for incentive integrity. The credit system requires only:*

1. *Local SNARK verification (computational soundness)*
2. *Local nullifier storage (bounded by $O(\text{credits_per_epoch})$ per verifier)*
3. *Bilateral receipt exchange (two-party protocol)*
4. *Per-epoch reputable set Merkle root (distributed, deterministic from penalty-log)*

Generality. Theorem 4 applies to any P2P system where: (a) work is bilateral, (b) credits are non-transferable and verifier-bound, (c) credit value exceeds production cost, and (d) work evidence is gated on partner reputation. This includes mesh networking relay, federated computation, and cooperative file storage.

5.5 Layer 1: Gossip Incentive Independence

Lemma 3 established that gossip is individually rational independent of relay incentives. We verify that the layered proof has no circular dependency:

1. Layer 0 (CRDT correctness): Axiomatic (set union).
2. Layer 1 (gossip): Depends only on non-transferability of credits (structural property, not on relay behavior).
3. Layer 2 (detection): Depends on gossip (Layer 1) + statistical threshold.
4. Layer 3 (relay): Depends on detection (Layer 2) + credit economics.

No circular dependency exists. Each layer’s proof assumptions are satisfied by layers below it, terminating at the axiomatic Layer 0.

6 Growth-Isolation Impossibility and Penalty-Log CRDT

Veil’s reputation system must function without consensus. We prove that this requires a careful separation of concerns, and that the separation is *necessary*, not merely convenient.

6.1 Impossibility Result

Theorem 6 (Growth-Isolation Impossibility). *Let $\mathcal{S}$ be a state-based CRDT maintaining a counter v that legitimately increases via verifiable operations (each operation is accompanied by unforgeable evidence e_i that can be independently validated). No merge function $m : \mathcal{S} \times \mathcal{S} \rightarrow \mathcal{S}$ simultaneously satisfies:*

1. **Inflation resistance:** *For any states S_A, S_B : $m(S_A, S_B).v \leq |\mathcal{E}(S_A) \cup \mathcal{E}(S_B)|$ where $\mathcal{E}(S)$ denotes the set of valid evidence contained in or referenced by state S .*
2. **Completeness:** *If node X has performed k verifiable operations with evidence $\{e_1, \dots, e_k\}$, and at least one of S_A, S_B has observed all k operations, then $m(S_A, S_B).v(X) \geq k$.*
3. **CRDT properties:** *m is commutative, associative, and idempotent.*

Proof. We prove this for all possible merge function strategies, including authenticated increment chains.

Case 1: Value-based merge (take max/min/average).

Let m be any function of the form $m(S_A, S_B).v = g(S_A.v, S_B.v)$ for some g .

- If $g(a, b) \geq \max(a, b)$: adversary creates S_A with $v = v_{\max}$. Merging with any state yields $\geq v_{\max}$. Inflation resistance violated.
- If $g(a, b) \leq \min(a, b)$: stale state S_A with $v = 0$ merged with current S_B yields ≤ 0 . Completeness violated.
- If $\min(a, b) < g(a, b) < \max(a, b)$ (e.g., average): stale state deflates current value. Completeness violated for any g that reduces the maximum.

Case 2: Evidence-set merge (authenticated increments).

Let m merge evidence sets: $m(S_A, S_B).v = |\mathcal{E}(S_A) \cup \mathcal{E}(S_B)|$. This trivially satisfies CRDT properties and completeness. We show it violates inflation resistance in an open network.

In an open network where evidence is transmitted between nodes, any valid evidence e_i observed by node Y can be *replayed* to node Z (who may not have seen it). This is not forgery— e_i is genuinely valid. The issue: evidence generated for interaction between nodes A and B is indistinguishable from evidence that B “should count toward B ’s reputation at all nodes.”

Specifically: adversary $\mathcal{A}$ observes k valid evidence items for honest node X (by interacting with X or observing others’ interactions). $\mathcal{A}$ creates state S_A containing all k evidence items. This is valid—the evidence IS genuine. Now $\mathcal{A}$ creates k' ADDITIONAL evidence items by self-interacting with Sybil nodes (fake relay operations). The merge cannot distinguish genuine evidence from Sybil-generated evidence without knowing which nodes are Sybils—which requires consensus on the Sybil set.

More precisely: inflation resistance requires $m(S_A, S_B).v \leq |\mathcal{E}_{\text{valid}}|$ where “valid” excludes Sybil-generated evidence. But determining which evidence is Sybil-generated requires knowing which nodes are Sybils, which IS the reputation problem being solved—circular.

Case 3: Vector clocks / causal history.

Vector clocks track per-node increments: $V[i]$ counts operations by node i . Merge: $m(V_A, V_B)[i] = \max(V_A[i], V_B[i])$. This satisfies CRDT properties. For reputation:

- Inflation resistance: requires each increment to $V[i]$ be authenticated by i . But an adversarial i can self-increment arbitrarily (the clock is self-reported). External validation requires knowing which operations are “legitimate”—again requiring reputation knowledge (circular).
- If increments require co-signatures (bilateral), we return to Case 2 (bilateral evidence sets) with the same Sybil problem.

Conclusion: No merge strategy can achieve inflation resistance + completeness + CRDT properties for growing reputation, because inflation resistance requires distinguishing Sybil-generated evidence from legitimate evidence, which requires exactly the reputation information being computed. $\square$ $\square$

6.2 Resolution: Penalty-Log CRDT

We resolve Theorem 6 by *not merging growth at all*. Growth is computed locally from verifiable evidence that the computing node has independently checked. Only penalties (negative events) are tracked in the CRDT.

Definition 14 (Penalty-Log CRDT). *State at node i :*

$$S_i = \{(n, P_n) : n \in V, P_n \subseteq \text{PenaltyEvent}\} \quad (6)$$

where $\text{PenaltyEvent} = (\text{event_id}, \text{evidence}, \text{epoch}, \text{severity})$.

Merge:

$$\text{Merge}(S_A, S_B) = \{(n, P_n^A \cup P_n^B) : n \in V\} \quad (7)$$

Reputation computation (local, deterministic given local observations):

$$R(n) = \underbrace{\left(1 - e^{-\lambda \cdot k_n}\right)}_{\text{growth from locally-verified proofs}} \times \underbrace{\prod_{p \in P_n} (1 - p.\text{severity})}_{\text{penalty factor from CRDT}} \quad (8)$$

where k_n is the number of valid relay proofs from node n that the local node has independently verified (NEVER merged—each node counts only what it has directly witnessed or verified via SNARK).

Proposition 7 (CRDT Correctness). *The Penalty-Log CRDT satisfies:*

1. **CRDT properties:** Set union is commutative, associative, idempotent.
2. **Inflation resistance:** Growth k_n is computed locally from independently-verified SNARK proofs. A Sybil cannot inflate k_n at node i without actually relaying packets that i observed (SNARK soundness). Penalty union can only decrease R .
3. **Completeness:** No stale state can decrease growth (growth is not merged). Valid old penalties remain valid and are included via union. A node’s local view of k_n may be an underestimate (not having observed all of n ’s relay work), but this is conservative, not inflationary.

Key insight: The impossibility theorem proves that inflation resistance and completeness CANNOT both hold for a merged value. The penalty-log CRDT sidesteps this by making growth *unmerged* (locally computed, possibly underestimated) and making penalties *merged* (conservatively overestimated—any penalty seen anywhere is applied everywhere). The asymmetry is deliberate: underestimating reputation is safe (conservative), while overestimating it enables attacks.

6.3 Applications Beyond Messaging

The penalty-log CRDT construction resolves a general problem: maintaining decentralized reputation without consensus. This applies to:

1. **Decentralized marketplaces:** Seller/buyer reputation without a central platform. Growth (successful transactions) is verified locally by interaction partners; penalties (fraud evidence) propagate globally.
2. **Distributed content moderation:** Community reputation for content curators without a central moderator. Locally-verified upvotes grow reputation; penalty events (rule violations with cryptographic evidence) propagate via CRDT.
3. **Peer-to-peer lending:** Creditworthiness without centralized credit bureaus. Repayment history is bilateral (verified by counterparty); defaults propagate as penalty events.
4. **Federated machine learning:** Node reliability scores for federated learning without a central coordinator. Contribution quality is verified locally by aggregators; model poisoning evidence propagates as penalties.

In each case, the same impossibility applies (reputation that grows via verifiable evidence cannot be merged without Sybil-vulnerability), and the same resolution works (don’t merge growth; merge only penalties).

6.4 Penalty Evidence and Detection

Definition 15 (Drop-Rate Penalty). *Node n is penalized when:*

$$\frac{\text{confirmed_drops}(n, W)}{\text{total_relayed}(n, W)} > \tau \quad (9)$$

over window W containing at least $W_{\min} = 500$ relay observations.

Proposition 8 (Framing Resistance). *An honest node’s observed drop rate due to adversarial framing concentrates around f^2 . The quadratic factor arises because a successful frame requires the adversary to control BOTH the node sending a packet TO the honest relay (who can claim the packet was sent) AND the next-hop node AFTER the honest relay (who can falsely claim non-receipt). Each occurs independently with probability f , giving framing rate $f \times f = f^2$. Setting $\tau > f^2$ ensures honest nodes are not falsely penalized.*

For $W_{\min} = 500$, $f = 0.1$, $\tau = 0.05$: The expected number of apparent drops is $W_{\min} \cdot f^2 = 5$ out of 500 observations. A false penalty requires observing $> \tau \cdot W_{\min} = 25$ apparent drops. By

the multiplicative Chernoff bound for $X \sim \text{Binomial}(500, 0.01)$:

$$P(X \geq 25) \leq \left(\frac{e^4}{5^5}\right)^5 < 10^{-20} \quad (10)$$

which is negligible for all practical security parameters.

Fundamental coupling: The maximum tolerable adversary fraction is bounded by $f_{\max} < \sqrt{\tau}$. For $\tau = 0.05$: $f_{\max} \approx 0.22$. This ensures the framing rate $f^2 < \tau$, maintaining negligible false-penalty probability.

7 Security Analysis

7.1 Sender Anonymity

Theorem 9 (Sender Anonymity). *Under the stratified mixnet with Poisson delays and cover traffic, Veil satisfies sender anonymity (Definition 1) against any PPT adversary $\mathcal{A}$ controlling $f < f_{\max}$ fraction of nodes, assuming:*

1. *The Decisional Diffie-Hellman (DDH) assumption holds over BabyJubjub (Sphinx indistinguishability).*
2. *The SNARK proof system satisfies zero-knowledge (credit proofs leak nothing).*
3. *The underlying Loopix mixing strategy provides sender anonymity against a GPA [8]. (Veil's GPA model subsumes Loopix's adversary model: Loopix assumes honest mix providers, which corresponds to honest relay nodes in Veil; adversarial nodes in Veil correspond to compromised providers in Loopix's analysis.)*

Proof. We reduce Veil's sender anonymity to Loopix's, showing that Veil's additions do not break the guarantee.

Reduction: Suppose $\mathcal{A}$ breaks Veil's sender anonymity with non-negligible advantage ϵ . We construct $\mathcal{A}'$ that breaks Loopix sender anonymity:

$\mathcal{A}'$ simulates the Veil protocol for $\mathcal{A}$ as follows:

1. Credit proofs are simulated using the SNARK simulator Sim (exists by zero-knowledge property). Simulated proofs are computationally indistinguishable from real proofs. Therefore credit proofs attached to messages provide $\mathcal{A}$ zero additional information about the sender.
2. Staggered share emission: Shares are emitted at the cover traffic Poisson rate λ_{cover} . By construction, the emission process for a sending node is statistically identical to its cover traffic process (a cover slot is replaced, not added). Therefore the emission pattern is identically distributed whether the node sends or not (Definition 3).
3. Adaptive cover rate: Cover traffic rate is bounded in $[\lambda_{\min}, \lambda_{\max}]$ for all nodes. The rate is a function of device state (battery, connectivity), NOT of sending behavior. Therefore $\mathcal{A}$ observing the rate learns device state (non-goal, acknowledged in threat model) but not sending behavior.
4. Nova folding state: The running IVC proof π_i is stored locally on the sender's device and is never transmitted over the network. The final compressed SNARK is pre-computed and cached after the N_{credit} -th fold (not generated at send time), so no network-observable computation correlates with message emission.

Since $\mathcal{A}'$'s simulation is computationally indistinguishable from real Veil execution (by SNARK zero-knowledge + DDH for Sphinx), any advantage $\mathcal{A}$ has against Veil translates to equal advantage against Loopix. By assumption (3), this advantage is negligible. $\square$ $\square$

Theorem 10 (Recipient Anonymity). *A global passive adversary cannot determine the recipient of a message with probability better than $1/|\mathcal{A}_R|$, where $\mathcal{A}_R$ is the set of all nodes performing fixed-interval mailbox retrievals during the delivery window.*

Proof. Fixed-interval padded retrieval ensures all recipients generate identical traffic patterns regardless of whether messages exist. Share deposit timing is decorrelated from retrieval timing by the storage delay at mailbox nodes. Formally: $\text{View}_{\mathcal{A}}(\text{recipient } r \text{ retrieves message})$ is identically distributed to $\text{View}_{\mathcal{A}}(\text{recipient } r \text{ retrieves padding})$ because both produce the same observable network event (fixed-size download at fixed interval). $\square$ $\square$

7.2 Constructive Adversary Property

We formalize the “constructive adversary” intuition with a quantitative entropy metric and rigorous probability bound.

Definition 16 (Sender Anonymity Entropy). *The sender anonymity entropy from the adversary’s perspective is:*

$$H(\text{Sender} \mid \text{Obs}_{\mathcal{A}}) = - \sum_{s \in V} P(S = s \mid \text{Obs}_{\mathcal{A}}) \cdot \log_2 P(S = s \mid \text{Obs}_{\mathcal{A}})$$

where $\text{Obs}_{\mathcal{A}}$ is the adversary’s complete observation (all network traffic + controlled node internal state).

Theorem 11 (Constructive Adversary Bound). *Let $\mathcal{A}$ control fraction f of relay nodes in a Veil network with N total nodes, L mix layers, threshold parameter T , and K total shares. Then:*

1. **Privacy provided** by $\mathcal{A}$ ’s nodes: *The conditional sender anonymity entropy satisfies:*

$$H(\text{Sender} \mid \text{Obs}_{\mathcal{A}}) \geq (1 - f^{L-1}) \cdot \log_2(f \cdot N) + f^{L-1} \cdot 0 \quad (11)$$

That is, with probability $1 - f^{L-1}$ (at least one non-adversarial hop exists on the path), the adversary’s uncertainty is at least $\log_2(f \cdot N)$ bits; only when ALL other hops are also adversarial (probability f^{L-1}) can uncertainty drop to zero.

2. **Privacy damaged** to any individual target: *The probability that $\mathcal{A}$ deanonymizes a specific sender is bounded by:*

$$P(\text{deanon}) \leq \binom{K}{T} \cdot f^{L \cdot T} \quad (12)$$

with a hypergeometric correction for finite layer sizes:

$$P(\text{deanon}) \leq \binom{K}{T} \cdot \prod_{j=1}^T \prod_{l=1}^L \frac{f \cdot N_l - (j-1)}{N_l - (j-1)} \quad (13)$$

For $L = 3$, $T = 3$, $K = 5$, $N_l \geq 1000$: $P(\text{deanon}) \leq 10 \cdot f^9 + O(1/N_l)$. For $f = 0.1$: $P(\text{deanon}) \leq 10 \cdot 10^{-9} = 10^{-8}$.

Since the expected entropy provided $\geq (1 - f^{L-1}) \cdot \log_2(f \cdot N)$ grows with N while the damage probability $\leq \binom{K}{T} \cdot f^{L \cdot T}$ is independent of N and exponentially small in $L \cdot T$, every adversary provides strictly more aggregate privacy than it can extract from any individual.

Proof. Part 1 (Privacy provided):

Each adversarial node in the mix layers processes packets for honest users (to earn credits and maintain reputation). Consider a packet traversing adversarial relay at layer l . The adversary observes the input/output of packets at its own nodes but cannot correlate across honest hops (by Sphinx indistinguishability under DDH).

Partition the adversary’s observation into two cases based on whether the remaining $L - 1$ hops contain any honest node:

Case A (probability $1 - f^{L-1}$): At least one other hop is honest. That hop applies Poisson mixing, making the adversary’s observation at its own node(s) statistically independent of the true sender. The adversary observes packets only at its own nodes; under uniform random routing, each adversarial node in layer l receives traffic from a fraction $1/N_l$ of all N senders. Across all $f \cdot N_l$ adversarial nodes in layer l , the adversary observes packets from $\approx f \cdot N$ distinct senders per mixing window. Its posterior is approximately uniform over these $f \cdot N$ candidates, giving uncertainty $\geq \log_2(f \cdot N)$ bits. (This assumes the mixing window contains $\Omega(f \cdot N)$ packets; at very low traffic in small networks the bound may be loose. Our empirical validation in §9 confirms it for $N \geq 500$.)

Case B (probability f^{L-1}): All other hops are also adversarial. The adversary controls the entire path and can potentially identify the sender with certainty (uncertainty = 0).

By the law of total expectation applied to entropy:

$$H(\text{Sender} \mid \text{Obs}_{\mathcal{A}}) \geq (1 - f^{L-1}) \cdot \log_2(f \cdot N) + f^{L-1} \cdot 0 = (1 - f^{L-1}) \cdot \log_2(f \cdot N)$$

For $f = 0.1$, $N = 1000$, $L = 3$: entropy $\geq (1 - 0.01) \cdot \log_2(100) \approx 0.99 \cdot 6.64 = 6.6$ bits.

Part 2 (Privacy damaged):

To deanonymize a specific target’s message, $\mathcal{A}$ must identify the sender of $\geq T$ shares (since $< T$ shares reveal nothing by information-theoretic security of Shamir sharing).

For each share j , the path $P_j = (v_{j,1}, v_{j,2}, v_{j,3})$ is selected by the sender with independent randomness: $v_{j,l}$ is chosen uniformly at random from layer l ’s node set of size N_l .

Independence justification: While all shares see the same layer composition (fixed per epoch), the path *selection* uses independent randomness per share. The selection is uniform within each layer, and shares do not coordinate their random choices. Therefore:

$$P(\text{all hops adversarial on path } j) = \prod_{l=1}^L \frac{|\mathcal{A} \cap \text{Layer}_l|}{|\text{Layer}_l|} = f^L$$

under uniform assignment of adversarial nodes across layers.

For independence across shares: shares use independent random coins for path selection. They sample from the same finite pools, but for $N_l \geq 1000$ and $K = 5$ shares, the probability of any node collision across shares is $\leq \binom{5}{2}/1000 = 0.01$. We include the exact hypergeometric correction for T shares all compromised:

$$P(\text{deanon}) \leq \binom{K}{T} \cdot \prod_{j=1}^T \prod_{l=1}^L \frac{f \cdot N_l - (j-1)}{N_l - (j-1)}$$

For large N_l : each factor $\approx f$, giving $\binom{K}{T} \cdot f^{L \cdot T}$. The factor $\binom{K}{T} = 10$ accounts for the adversary choosing which T of K shares to target.

Part 3 (Constructive property):

We demonstrate two independent properties:

1. Adversarial participation necessarily provides substantial entropy to honest users: $(1 - 0.01) \cdot \log_2(100) \approx 6.6$ bits for $f = 0.1$, $N = 1000$.
2. The deanonymization probability per target is negligible: $10 \cdot 10^{-9} = 10^{-8}$ for the same parameters.

These quantities—Shannon entropy (information-theoretic, aggregate) and deanonymization probability (frequentist, per-target)—are not directly commensurable. We do not claim a formal utility comparison between them. Rather, we observe that adversarial presence simultaneously (a) provides a measurable privacy contribution that cannot be withheld (by the relay work requirement) and (b) enables only negligible per-target damage. A formal comparison across these incommensurable quantities would require a privacy utility framework, which we leave to future work. □ □

Corollary 12 (Attack Economics). *For an adversary to achieve deanonymization probability p against a specific target, they must control $f = (p/\binom{K}{T})^{1/(L \cdot T)}$ fraction of reputable nodes. At $L = 3, T = 3, K = 5$: achieving $p = 0.01$ requires $f = (0.001)^{1/9} \approx 0.50$ —controlling 50% of all reputable nodes.*

Under the social Sybil resistance mechanism (Proposition 2), achieving $f = 0.50$ in a 10,000-node network requires $\sim 5,000$ Sybil nodes. The time required scales inversely with the number of seed nodes s the adversary controls: $\text{time} = 5000/(s \times B_{\max})$ credit windows. For $B_{\max} = 3$ vouches per credit window (24 days):

Seeds (s)	Credit windows	Time
10	167	~ 11 years
50	33	~ 2.2 years
100	17	~ 1.1 years

However, each seed must itself be reputable ($R > R_{\min}$), requiring prior honest relay work. Acquiring 100 reputable seeds already demands significant honest contribution to the network. Furthermore, during the entire accumulation period, all Sybils must relay honestly (providing genuine mixing service), consuming real computational and bandwidth resources—a direct instantiation of the constructive adversary property.

7.3 Spam Resistance

Proposition 13 (Bounded Amplification). *Sending M messages requires at least $M \cdot N_{\text{credit}}$ honest relay operations. The amplification factor is exactly 1: each credit is usable at exactly one verifier (Proposition 1).*

Proof. Each message requires a fresh SNARK proof covering N_{credit} receipts with a verifier-bound nullifier. The same proof cannot be reused at a different first-hop (verifier-binding). Receipts are bound to the sender’s secret key (non-transferable). Receipt issuers must be in the reputable set (no Sybil self-relay). Therefore each message consumes exactly N_{credit} distinct relay operations, each performed for a genuinely reputable counterpart. $\square$ $\square$

7.4 Liveness

Proposition 14 (Delivery Guarantee). *A message is delivered within 7 days of sending with probability $\geq 1 - (1 - q)^K$, where q is the probability that a single share reaches its mailbox and persists until retrieval. For $K = 5$ and $q \geq 0.85$: delivery probability $\geq 1 - 0.15^5 > 0.99999$.*

8 Feasibility Analysis

8.1 Computational Cost

Derivation: A full-rate node relays $\approx 0.65 \times 10,800 \approx 7,000$ packets per 3-hour active session (rate from credit window calibration above). Then $7,000 \times 150\text{ms} = 1,050\text{s} \approx 17.5$ minutes. At reduced rate ($\lambda/4$): $1,750 \times 150\text{ms} = 262\text{s} \approx 4.4$ minutes. Timing is extrapolated from the measured 12,387 constraints (Table 1), assuming linear scaling from the 200ms/16,400-constraint baseline. These computations occur incrementally during relay operations (each fold happens immediately after relay, not batched), so peak CPU load is one 150ms burst per relay event ($\approx$ every 12 seconds at full rate). Actual hardware benchmarks on ARM Cortex-A53 are pending.

Table 2: Proving costs on mid-range smartphone (2025 hardware, extrapolated from measured constraints)

Operation	Constraints	Estimated Time
Single relay fold (Nova)	12,387	~150ms
Merkle proof per receipt	3,403	(included above)
Final SNARK compression	—	~1.5s
Daily relay proving (7,000 pkts)	—	~17.5 min
Daily relay proving (adaptive $\lambda/4$)	—	~4.4 min

Table 3: Daily bandwidth per node

Traffic type	Volume	Derivation
Cover traffic (full rate)	38.4 MB/day	$0.217 \times 86400 \times 2048$ bytes
Cover traffic (adaptive $\lambda/4$)	9.6 MB/day	$0.054 \times 86400 \times 2048$ bytes
Relay forwarding	~14 MB/day	Proportional to layer load
Gossip (penalty-log CRDT)	< 1 MB/day	~200 bytes/event, rare

8.2 Bandwidth

8.3 Storage

- **Penalty-log CRDT:** ~200 bytes per penalty event. At 1,000 penalties/year in a 10,000-node network: ~200 KB/year.
- **Nullifier storage** (per verifier node): Each nullifier is 32 bytes. At ~1,000 messages verified per epoch: 32 KB/day, cleared each epoch.
- **Reputable set Merkle tree:** One root per epoch (32 bytes). Full tree for membership proofs: $N \times 32$ bytes $\approx$ 320 KB for $N = 10,000$.

8.4 Latency

Expected message delivery latency:

- Share emission (5 shares at cover rate $\sim 1/4.6$ s): ~23 seconds
- Mixing delay per share (3 hops): ~2–6 seconds
- Total: 30–80 seconds typical, up to 120 seconds at 99th percentile

8.5 Accessibility: Low-End Device Requirements

For Veil to serve its target population (users under mass surveillance in the Global South), it must function on sub-\$100 Android devices. We analyze minimum requirements:

Table 4: Minimum device requirements vs. representative low-end hardware

Resource	Veil Requirement	Typical Sub-\$100 Device (2025)
RAM	~50 MB (circuit + witness)	2–3 GB
Storage	~5 MB (state + proofs)	32–64 GB
CPU	200ms burst per relay (ARM Cortex-A53)	Quad-core A53 or better
Bandwidth	9.6 MB/day (adaptive mode)	500 MB–2 GB/month plans
Battery	~5.8 min CPU/day at $\lambda/4$	3000–5000 mAh typical

At adaptive rate ($\lambda/4$), total daily bandwidth is: cover traffic (9.6 MB sending) + relay

obligations ($\sim 3\text{--}7$ MB receiving, depending on layer assignment for a mobile node with reduced uptime) + gossip (< 1 MB) $\approx 15\text{--}18$ MB/day $\approx 450\text{--}540$ MB/month. This fits within common prepaid data plans in the Global South ($\$1\text{--}3$ /month for 1 GB in India, Kenya, Bangladesh), though users on the cheapest plans (< 500 MB/month) would need WiFi supplementation. The 200ms per-fold proving time is achievable on ARM Cortex-A53 cores (verified against arkworks-rs benchmarks for BN254 scalar multiplication on comparable hardware).

9 Empirical Validation

We validate the key theoretical bounds via Monte Carlo simulation. Source code is available in the supplementary materials.

9.1 Constructive Adversary Bound (Theorem 11)

We simulate a 5,000-node Veil network ($L = 3$, $K = 5$, $T = 3$, $N_l = 1,666$ per layer) under varying adversary fractions $f \in \{0.05, 0.08, 0.10, \dots, 0.22\}$. For each configuration, we run up to 10 million message-send trials, measuring the empirical deanonymization rate (fraction of messages where the adversary controls all L hops on $\geq T$ share paths).

Table 5: Monte Carlo validation of Theorem 11 ($N = 5,000$, up to 10M trials per f)

f	Trials	Empirical $P(\text{deanon})$	Theoretical bound	Status
0.05	500K	$< 2 \times 10^{-6}$ (0 events)	1.70×10^{-11}	Consistent
0.10	500K	$< 2 \times 10^{-6}$ (0 events)	9.22×10^{-9}	Consistent
0.15	10M	4.0×10^{-7} (4 events)	3.85×10^{-7}	Consistent [†]
0.18	10M	1.7×10^{-6}	1.88×10^{-6}	Holds (ratio 0.90)
0.20	10M	4.9×10^{-6}	4.98×10^{-6}	Holds (ratio 0.98)
0.22	10M	1.11×10^{-5}	1.17×10^{-5}	Holds (ratio 0.95)

[†] 4 events in 10M trials (expected $\lambda = 3.85$); Poisson $P(X \geq 4 \mid \lambda = 3.85) = 0.47$ — cannot reject bound at any reasonable significance level.

Results: At all tested adversary fractions, the empirical deanonymization rate is consistent with the theoretical bound $\binom{K}{T} \cdot f^{L \cdot T}$. For $f \geq 0.18$ (where sufficient events occur for statistical comparison), the empirical-to-theoretical ratio lies in $[0.90, 0.98]$, confirming the bound is tight. For $f < 0.15$, the theoretical bound implies fewer than 2 expected events per 5M trials; validation at these levels relies on the analytical proof in Theorem 11 rather than empirical measurement. The constructive adversary property is verified: entropy provided ≥ 7.95 bits across all f values, while deanonymization probability remains $\leq 1.11 \times 10^{-5}$ even at the maximum tolerable $f = 0.22$.

9.2 Implementation Validation

We implement the core protocol in 2,857 lines of Rust (Apache 2.0), including the relay proof circuit, penalty-log CRDT, credit state machine, Sphinx packet processing, and Shamir secret sharing. All 53 tests pass (34 protocol correctness + 9 circuit + 10 adversarial-input security tests), verifying:

- **Circuit correctness:** Relay step circuit (12,387 constraints on arkworks 0.4 [34]) accepts honest witnesses and rejects forged receipts, corrupted Merkle proofs, and invalid issuers.
- **CRDT properties:** Commutativity, associativity, and idempotency verified across 50 random merge orderings—reputation values are identical regardless of event arrival order.
- **Nullifier binding:** Same credit proof produces different nullifiers for different verifiers, preventing cross-verifier reuse without consensus.

- **Shamir correctness:** All $\binom{5}{3} = 10$ subsets of 3-from-5 shares correctly reconstruct; fewer than 3 shares fail as required.
- **Cover traffic:** Exponential inter-arrival times match configured rates within 2%; mobile mode ($\lambda/4$) verified $4\times$ slower; real shares replace cover slots (not additive).
- **Adversarial inputs:** Security tests validate rejection of forged receipts, non-reputable issuers, stale-epoch Merkle roots, self-relay attempts (duplicate issuers), cross-verifier nullifier reuse, insufficient relay work ($< N_{\text{credit}}$ receipts), and CRDT growth inflation attacks.

Network size sensitivity: At $f = 0.10$, the bound holds identically for $N \in \{500, 1,000, 5,000, 50,000\}$, confirming that the hypergeometric correction is negligible for $N_l \geq 166$ (all tested sizes).

9.3 Penalty-Log CRDT Convergence

We simulate epidemic gossip propagation of penalty events in a 5,000-node network with fanout $k = 8$ (each node gossips to 8 random peers per round).

Table 6: CRDT convergence: rounds to 95% propagation under adversarial conditions

Condition	Mean rounds	Max rounds
Baseline (no adversary)	5.0	5
5% churn (nodes offline/round)	5.0	5
10% adversarial withholding	5.0	5
Combined (5% churn + 10% withholding)	5.0	5
Maximum adversary ($f = 0.22$ withholding + 5% churn)	5.0	6

Results: Penalty events reach 95% of honest online nodes within 5–6 gossip rounds regardless of adversarial conditions, up to the maximum tolerable $f_{\text{max}} = 0.22$. Convergence scales as $O(\log N)$: at $N = 100$ convergence takes 3 rounds; at $N = 10,000$ it takes 5 rounds. Adversarial withholding delays propagation by at most 1 round due to redundant gossip paths.

Merge order independence: We verify that 50 simulated observers receiving the same penalty events in random permutations compute identical reputation values, confirming the CRDT commutativity guarantee (Proposition 7).

10 Limitations and Open Problems

We explicitly state what Veil does *not* protect against and what remains unresolved.

10.1 Fundamental Limitations

1. **Patient Sybils.** Any reputation system based on observed behavior cannot distinguish patient adversaries from genuine users. A Sybil that relays honestly for years is behaviorally identical to an honest node. Our defense: such Sybils contribute genuine relay work during their building phase (the constructive adversary property—Theorem 11). The attack cost is measured in node-years of honest service. Upon activation, reputation contagion rapidly degrades the adversary’s infrastructure.
2. **Subtle QoS Degradation.** Nodes dropping fewer than τ fraction of packets go undetected. Defense: threshold secret sharing absorbs up to $K - T = 2$ share losses per message. At maximum undetectable dropping ($\tau = 0.05$) across 3 layers, effective share loss rate is $\leq 1 - (1 - 0.05)^3 \approx 0.14$. Message failure rate: $(0.14)^3 / \binom{5}{3} \approx 0.03\%$ (negligible).
3. **Penalty Irreversibility.** The grow-only penalty set means erroneous penalties cannot be removed. However, the false-penalty probability is $< 10^{-20}$ per node per window (Propo-

sition 8). Over a 10,000-node network running 10 years with daily windows: expected false penalties $< 10^{-20} \times 10^4 \times 3650 \approx 10^{-13}$.

4. **Traffic Analysis Under ML.** Recent work [18] demonstrates ML-based traffic analysis can correlate flows even through Poisson-delayed mixnets. Veil’s cover traffic mitigates but does not provably prevent this. Formal resistance against adaptive ML adversaries remains an open problem for ALL mixnet architectures.
5. **Mobile Participation Asymmetry.** Mobile nodes with adaptive cover traffic rates ($\lambda/4$) contribute less anonymity than always-on nodes. The anonymity set is dominated by high-uptime nodes; mobile participants benefit from this set without fully contributing.
6. **Bootstrap Phase Privacy.** Nodes in the training layer (Layer 0) receive weaker anonymity guarantees due to the smaller anonymity set.
7. **Collusive Credit Generation.** If k reputable nodes collude, they can issue receipts to each other without relaying genuine traffic (“wash-relaying”). The circuit cannot distinguish colluded receipts from legitimate ones because the privacy guarantee prevents verifiers from learning which specific packets were relayed. Defense: (a) becoming reputable requires social vouching ($K_{\text{vouch}} = 3$ vouchers per node, depth-limited, linear growth), creating irreducible time-cost for establishing a colluding ring; (b) colluding nodes must maintain relay assignments to remain reputable, meaning they still process honest traffic (contributing genuine mixing); (c) the damage is bounded: k colluding nodes can generate at most $k \times (k - 1)$ fake receipts per epoch (each pair generates one), while honest traffic generates $\gg k^2$ legitimate receipts in any network with $N \gg k$. The spam amplification from collusion is $O(k^2/N)$ — negligible for small k , and acquiring large k requires the social attestation time-cost. This is the fundamental tension: stronger privacy for relay proofs enables collusion, while weaker privacy would create metadata leakage. We accept bounded collusion as the cost of strong relay privacy.
8. **Composition Security.** Veil composes Sphinx, Poseidon, Nova, Groth16, CRDTs, and Shamir secret sharing. Our sender anonymity proof (Theorem 9) reduces to Loopix under component-level assumptions (DDH, SNARK zero-knowledge). However, we do not provide a Universal Composability (UC) proof of the full protocol. Potential composition concerns include: (a) whether Nova folding state leaks side-channel information under implementation, (b) whether the epoch-synchronized Merkle tree update creates timing fingerprints, and (c) whether the interaction between credit verification timing and routing enables intersection attacks. We believe these are implementation-level concerns (not protocol-level), but formal UC analysis is required to confirm this.

10.2 Censorship Resistance

Veil’s threat model assumes network connectivity between nodes but does not explicitly address ISP-level protocol blocking. In jurisdictions where state-level deep packet inspection (DPI) is deployed (China, Iran, Russia), Veil traffic may be identifiable and blocked.

Mitigation approaches (not implemented in this work, but compatible with the architecture):

- **Pluggable transports:** Veil’s fixed-size packets can be encapsulated within obfuscated transport layers (e.g., obfs4 [32], Snowflake [33]) that make traffic indistinguishable from allowed protocols (HTTPS, WebSocket, video streaming).
- **Bridge nodes:** A subset of Veil nodes with public-facing HTTPS endpoints can serve as entry points for users behind DPI, analogous to Tor bridges. Bridge addresses are distributed out-of-band.

- **Domain fronting:** Where available, Veil traffic can be tunneled through CDN infrastructure such that the true destination is hidden from network observers.

Veil’s architecture is transport-agnostic by design: the relay proof layer operates on fixed-size packets regardless of how they are transported between nodes. Pluggable transports compose without modification to the credit system, penalty-log CRDT, or mixing logic. A formal analysis of Veil’s resistance to active traffic manipulation (injection, replay, selective blocking) under censorship is left to future work.

10.3 Ethical Considerations and Dual-Use

Any privacy-preserving communication system faces the dual-use challenge: the same properties that protect journalists and dissidents also benefit criminals. We address this directly.

What Veil does NOT prevent:

- Recipients can still report message content to authorities (content is decrypted at the endpoint).
- Community-level moderation is possible: recipients can maintain local blocklists, and communities can establish shared penalty events for users who violate agreed norms.
- Law enforcement with lawful access to an endpoint device obtains all decrypted messages (Veil protects metadata in transit, not data at rest on compromised devices).

Design choices reflecting ethical constraints:

- The spam prevention mechanism (relay work requirement) limits bulk distribution of harmful content.
- The social vouching requirement creates accountability chains—misbehavior affects vouchers via reputation contagion, incentivizing community self-governance.
- High-latency design (30–120 seconds) makes Veil unsuitable for real-time coordination of time-sensitive criminal activity while remaining fully functional for the journalist, activist, and whistleblower use cases.

We adopt the ethical framework established by the Tor Project [31]: privacy tools should be evaluated by whether they advance human rights on balance, not by whether any individual can misuse them. The alternative—communications without metadata protection—enables mass surveillance that demonstrably harms billions, while targeted law enforcement remains possible via endpoint access.

10.4 Open Problems

1. Formal Universal Composability (UC) proof of the full protocol composition. We intend to pursue machine-checked verification of Theorems 4 and 6 in Lean 4, providing the highest level of confidence in correctness.
2. Optimal cover traffic rates minimizing bandwidth while maintaining provable unobservability against polynomial-time adversaries.
3. Post-quantum migration path (BabyJubjub/BN254 are not post-quantum secure; lattice-based SNARKs are an active research direction).
4. Extension to real-time messaging via reduced mixing delays with formal characterization of the privacy-latency trade-off.
5. Formal analysis of ML traffic resistance: characterize the class of ML adversaries against which Poisson mixing + cover traffic provides provable guarantees.
6. Pluggable transport integration for censorship-resistant deployment (§10.2).

11 Toward Deployment

For Veil to achieve its societal purpose—providing metadata-private communication to populations under mass surveillance—the protocol must progress from theory to deployed system. We outline the path.

11.1 Reference Implementation Roadmap

1. **Phase 1: Core Cryptography** (Months 1–3). Relay proof circuit implementation in arkworks-rs (Rust). Nova folding integration via nova-snark crate. Benchmark validation on ARM Cortex-A53 hardware to confirm Table 2 estimates. Deliverable: open-source circuit library with test vectors.
2. **Phase 2: Networking Layer** (Months 4–6). Penalty-log CRDT gossip protocol over libp2p. Sphinx packet construction and routing. Basic CLI client supporting send/receive. Deliverable: functional testnet (100+ nodes on commodity hardware).
3. **Phase 3: Mobile Client** (Months 7–12). Cross-platform mobile application (Rust core via FFI, native UI). Adaptive cover traffic mode. Bootstrap protocol integration. Battery and bandwidth optimization for sustained background operation. Deliverable: beta-quality mobile client.
4. **Phase 4: Production Hardening** (Months 12–18). Security audit by independent cryptography firm. Pluggable transport integration for censorship resistance. Deployment partnerships with press freedom organizations. Deliverable: production-ready system with initial user base.

Resourcing assumption: This timeline assumes 2–3 full-time engineers with cryptographic systems experience. For a single researcher, phases extend proportionally (estimated 24–30 months to Phase 3 completion).

11.2 Deployment Strategy

We envision a staged deployment reflecting the sensitivity of the target use cases:

1. **Controlled pilot** (Year 1): Partnership with 2–3 press freedom organizations (e.g., Committee to Protect Journalists, Reporters Without Borders) for journalist-source communication in high-risk environments. Small network (~500 nodes), monitored for protocol correctness.
2. **Community expansion** (Year 2): Organic growth via civil society networks. Open-source release enabling independent deployment. Target: 5,000–10,000 nodes across 20+ countries.
3. **General availability** (Year 3+): Public release with user-friendly onboarding. Integration with existing secure communication workflows.

11.3 Governance

Protocol parameter changes (τ , $f_{\max}$, N_{credit} , W_{credit}) and upgrades must be coordinated without central authority. We propose:

- **Rough consensus:** Parameter proposals are published as signed documents in the penalty-log CRDT. Nodes signal support by including a commitment to the proposal in their next penalty-log gossip round. A proposal activates when $> 2/3$ of reputable nodes (by reputation-weight) signal support within a coordination window.

- **Fork freedom:** Any node may reject a parameter change by continuing to operate under old parameters. The protocol degrades gracefully under parameter mismatch (credits remain locally verifiable regardless of global parameter agreement).
- **Licensing:** The reference implementation and protocol specification will be released under the Apache 2.0 license, ensuring unrestricted use, modification, and deployment by any individual or organization.

11.4 Quantitative System Comparison

Table 7: Quantitative comparison across key dimensions

Metric	Veil	Nym	Tor	Loopix	Pung
Latency (typical)	30–80s	1–5s	0.5–2s	2–10s	10–30s
Bandwidth/user/day	10–38 MB	5–20 MB	1–5 MB	20–50 MB	50+ MB
Economic requirement	None	NYM token	None	None	None
Blockchain required	No	Yes	No	No	No
Trusted infrastructure	None	Validators	Dir. auth.	Providers	Servers
Mobile feasible	Yes	Partial	No	No	No
Spam resistance	Built-in	Token stake	None	None	Computational
Offline delivery	7 days	No	No	Partial	No
Adversary model	$GPA + f < 0.22$	$GPA + \text{stake}$	Non-global	GPA	GPA
Anonymity set	N nodes	N nodes	Circuit-based	N nodes	N users

Veil’s primary trade-off is latency: 30–80 seconds versus sub-second for Tor. This is inherent to the security model (Poisson mixing + threshold delivery requires time dispersion for unlinkability). For the target use cases (journalist-source communication, activist coordination, whistleblowing), this latency is acceptable—these are asynchronous communication patterns where minutes of delay are routine.

12 Related Work

Table 8: Feature comparison with existing systems

System	Metadata	Offline	Spam	Decentralized	No Token	Mobile
Signal [2]		✓	Centralized		✓	✓
Nym [4]	✓		Token stake	✓		Partial
Tor [5]	Partial			✓	✓	
Loopix [8]	✓	Partial		Partial	✓	
SimpleX [6]	Partial	✓		Partial	✓	✓
HOPR [7]	✓		Token stake	✓		
Katzenpost [12]	✓	✓		Partial	✓	
Karaoke [23]	✓				✓	
Pung [24]	✓				✓	
Stadium [25]	✓			Partial	✓	
Veil	✓	✓	Relay work	✓	✓	✓

Incentivized relay networks. Nym [4] and HOPR [7] incentivize relay operators via cryptocurrency staking and rewards. HOPR uses the term “Proof of Relay” for its on-chain verification that a relay hop was completed, rewarding relayers with HOPR tokens via payment

channels on Ethereum. Veil’s Proof-of-Relay is fundamentally different in mechanism: HOPR requires token staking, blockchain payment channels, and on-chain settlement (creating economic barriers and metadata leakage from publicly visible transactions); Veil uses non-transferable bilateral credits verified locally via zero-knowledge proofs, requiring no blockchain, no tokens, and no global state. The distinction is not merely engineering but theoretical: Veil’s Theorem 4 proves that verifier-bound non-transferable credits achieve incentive compatibility *without consensus*—eliminating the double-spending problem structurally rather than solving it via a ledger. Nym similarly requires a globally-agreed token balance (consensus); Veil requires only local SNARK verification.

Metadata-hiding communication. Karaoke [23] achieves metadata-hiding via dialing protocols with cover traffic but requires coordinated servers. Pung [24] uses private information retrieval (PIR) for metadata privacy but has $O(\sqrt{N})$ server computation per message, limiting scalability. Stadium [25] scales metadata-hiding via parallel mixnet chains with verifiable shuffles but requires trusted setup among servers. Atom [26] achieves horizontal scaling via partitioned anonymity sets. Unlike these systems, Veil requires no trusted servers, no PIR infrastructure, and operates on commodity mobile hardware with decentralized coordination.

Social Sybil resistance. SybilGuard [13] and SybilLimit [14] use social graph random walks for Sybil detection. These systems operate on the social graph directly (revealing relationships) and do not integrate with anonymous communication. Veil’s Proof-of-Entanglement verifies social attestation in zero knowledge and couples it to relay incentives, preventing the social graph from becoming a metadata source.

Rate limiting nullifiers. RLN [20] provides Sybil-resistant rate limiting via nullifiers in a group signature scheme, primarily designed for Ethereum’s consensus layer. Veil’s nullifier design differs in two respects: (1) Veil nullifiers are verifier-bound (preventing cross-verifier reuse without consensus), while RLN nullifiers require a shared membership set; (2) Veil couples nullifiers to relay work evidence rather than membership proofs alone, creating a productive Sybil cost.

CRDTs for distributed state. Shapiro et al. [15] provide the foundational framework for conflict-free replicated data types. To our knowledge, no prior work applies CRDTs to reputation in anonymous relay networks, nor formalizes the growth-isolation impossibility for monotonically growing verifiable evidence. The closest work is Hellas [28] (peer-to-peer reputation with gossip), which assumes honest majority in gossip—an assumption our Lemma 3 removes.

Tor incentive proposals. BRAIDS [16], LIRA [17], and Gold Star [27] use transferable credits or differentiated service classes to incentivize Tor relays. All face Sybil attacks because credits can be earned by cheap adversarial relays and potentially transferred. Veil’s non-transferable, reputation-gated credits eliminate this attack vector: earning credits requires relaying for reputable nodes, which requires social vouching to join, creating irreducible time-cost.

Session. Session [35] (formerly Loki) provides decentralized messaging via onion-routed requests through incentivized service nodes. However, service node operation requires staking cryptocurrency (OXEN tokens), reintroducing the economic barrier that Veil eliminates. Session’s incentive model also permits token-based Sybil attacks: an adversary with sufficient capital can operate multiple service nodes without social cost.

13 Conclusion

We have presented Veil, a decentralized messaging protocol that solves four coupled problems—metadata resistance, offline delivery, spam prevention, and Sybil resistance—through a single unified mechanism: Proof-of-Relay. Our protocol requires no cryptocurrency, no blockchain, no trusted third parties, and no economic investment. Privacy is earned through device contribution alone.

Our theoretical contributions—the bilateral credit incentive theorem (with layered non-circular proof and verifier-bound nullifiers), the growth-isolation impossibility (covering all merge strategies including authenticated increments), and the constructive adversary bound (with formal entropy quantification and hypergeometric correction)—generalize beyond messaging to any peer-to-peer system requiring fair exchange without central coordination.

We propose the *constructive adversary principle* as a design heuristic for future decentralized systems: any system where adversarial participation requires productive work for honest users provides inherent resistance proportional to the adversary’s investment. We have demonstrated this principle formally in the context of mixnet relay; empirical validation across other system types (federated learning, decentralized storage, mesh networking) remains future work.

Veil demonstrates that the four properties long considered to require separate mechanisms (anonymity, reliability, fairness, decentralization) can emerge from a single design principle: *making the work that earns the right to send identical to the work that provides privacy to others*.

According to the World Bank Global Findex [30], 1.4 billion adults remain unbanked as of 2024—excluded from any system requiring financial infrastructure. An estimated 4.5 billion people live under governments conducting mass surveillance [29]. For these populations, cryptocurrency is often banned, surveilled, or simply inaccessible. Veil is the first metadata-private messaging system accessible to this population: it requires only a phone and intermittent connectivity. The path from theory to deployed infrastructure (§11) is the next step toward communication systems that serve all people regardless of economic means or jurisdictional constraints.

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